

EXPERIMENT 2

For a given set of training data examples stored in a .CSV file, implement and demonstrate the

Candidate-Elimination algorithm in python to output a description of the set of all hypotheses

consistent with the training examples

AIM

To implement and demonstrate the Candidate-Elimination Algorithm in Python for a given set of training examples stored in a CSV file and to find the set of all hypotheses consistent with the training examples.

PROCEDURE

Import the required Python libraries.

Create or load the training dataset from a CSV file.

Separate the attributes and target class.

Initialize the Specific Hypothesis (S) with the most specific hypothesis.

Initialize the General Hypothesis (G) with the most general hypothesis.

For every positive training example, generalize the specific hypothesis.

For every negative training example, specialize the general hypothesis.

Update the specific and general hypothesis boundaries.

Display the final hypothesis boundaries.

CODE:

```

import pandas as pd

data = [
["Sunny", "Warm", "Normal", "Strong", "Warm", "Same", "Yes"],
["Sunny", "Warm", "High", "Strong", "Warm", "Same", "Yes"],
["Rainy", "Cold", "High", "Strong", "Warm", "Change", "No"],
["Sunny", "Warm", "High", "Strong", "Cool", "Change", "Yes"]
]

columns = ["Sky", "AirTemp", "Humidity", "Wind", "Water", "Forecast", "EnjoySport"]

df = pd.DataFrame(data, columns=columns)

df.to_csv("training.csv", index=False)

print("CSV File Created Successfully")
print(df)

```

```

import pandas as pd

df=pd.read_csv("training.csv")
X=df.iloc[:, :-1]
Y=df.iloc[:, -1]

P=X[Y=="Yes"]

S=[P.iloc[0,i] if P.iloc[:,i].nunique()==1 else "?" for i in range(X.shape[1])]

N=X[Y=="No"]

G=[[S[j] if j==i else "?" for j in range(len(S))] for i in range(len(S))]

G=[g for g in G if any(g[j]!="?" and all(N.iloc[:,j]!=g[j]) for j in range(len(S)))]

print("Specific Hypothesis:")
print(S)

print("\nGeneral Hypothesis:")
print(G)

```

OUTPUT:

CSV File Created Successfully

	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes

Specific Hypothesis:

['Sunny', 'Warm', '?', 'Strong', '?', '?']

General Hypothesis:

[['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?']]

RESULT

The Candidate-Elimination Algorithm was successfully implemented and demonstrated using Python. The training data was stored and read from a CSV file.

The Specific Hypothesis (S) and General Hypothesis (G) were obtained, representing the boundaries of the Version Space containing hypotheses consistent with the given training examples